

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 30%

Annexure A

TECHNICAL PRESENTATION

Code of Conduct:

I R. MANOJ KUMAR NAIK (192524436) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

R. MANOJ KUMAR NAIK

Annexure A

TECHNICAL PRESENTATION

1. Role of the OSI Model in Software-Defined Networking (SDN)
2. Network Function Virtualization (NFV) and the OSI Architecture
3. Artificial Intelligence for Network Traffic Management
4. Machine Learning-Based Error Detection in Networks
5. Digital Twin Networks and the OSI Model
6. 6G Communication Architecture Compared with the OSI Model
7. Edge Computing and Reliable Data Transmission
8. Cloud Networking Through the OSI Architecture
9. Satellite Internet Communication Using OSI Layers
10. Future of Reliable Networking Beyond the Traditional OSI Model

RUBRICS FOR CALCULATION

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate	Good analysis	Basic analysis	Poor or incorrect

		interpretation and validation of results	with reasonable interpretation	with limited insights	analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

ARTIFICIAL INTELLIGENCE FOR NETWORK TRAFFIC MANAGEMENT

CSA0716-COMPUTER NETWORKS

UNDER THE GUIDANCE OF :
Dr.P.ABERNA

PRESENTED BY:
R.MANOJ KUMAR NAIK(192524436)
P.SRI KOUSICK(192524431)



INTRODUCTION



Artificial Intelligence for Network Traffic Management

- AI improves the efficiency of modern computer networks.
- It monitors, analyses, and controls network traffic using machine learning algorithms.
- Widely used in data centres, automatically.
- AI helps reduce congestion and improves data transmission speed.
- It supports real-time decision-making loud computing, and 5G networks.

PROBLEM STATEMENT

- Challenges in Traditional Network Traffic Management
- Increasing network traffic causes congestion.
- Manual traffic management is slow and inefficient.
- Difficult to predict sudden traffic spikes.
- High latency and packet loss reduce network performance.
- Traditional methods cannot adapt quickly to changing network conditions.

Need for AI in Network Traffic Management

Why AI is Needed?

Rapid growth of internet users and connected devices.

Increasing demand for high-speed and reliable communication.

Dynamic traffic patterns require adaptive management.

Manual monitoring is time-consuming and prone to errors.

AI enables proactive traffic control and congestion prevention.

